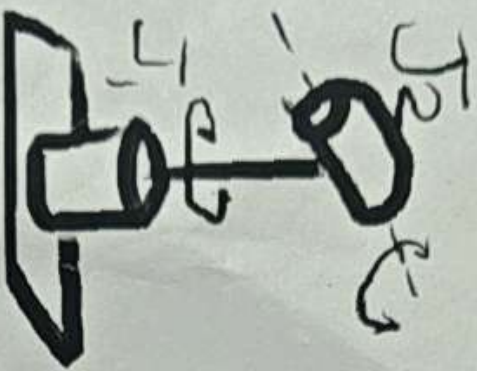


1 ✓ A mobile frame of reference initially coincides with a fixed frame. It is then rotated by 30° about the y-axis of the fixed frame. The mobile frame is then translated by 4 units along its own x-axis (new). If the coordinates of a point P with respect to the fixed frame are (10, 20, 40), find its coordinates with respect to the mobile frame. (5)

2



For the part of robot arm shown on left, assign the link frames L_0 and L_1 using D-H representation. Also list the four kinematic parameters for $k=1$. (Assume suitably, if required) (5)

3 ✓ The coordinates of the far end of Link-2 in a 2R planar manipulator with respect to the base frame are (173.2, 200). If the lengths of Links-1 and 2 are 100 mm and 200 mm, respectively, find the joint angles for "Elbow-down" configuration. (5)